Automating Suricata rule implementations

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1 Introduction

This paper presents a workflow on how to automate the process of implementing rules using the intrusion detection system Suricata.

2 Methodology

The framework is divided into four different activities to provide a comprehensive and structured overview. This process can be visualized in Figure 1.

2.1 Activity 1

The first activity is to fetch information from open-source intelligence datasets. In this activity, the data is unpacked and validated to create a lookup table of services requiring rule implementation. Please see `docs/inventorylist.pdf` for more information.

2.2 Activity 2

The second activity is to identify rules for `suricata_custom_rules_all.rules` and environment specific rules `suricata_custom_rules_x.rules` where `x` identifies an environment. As indicated, this is iterated as long as there are elements in the `lookup table` to fetch.

2.3 Activity 3

The third activity is to build rule files for `suricata_custom_rules_all.rules` and for environment-specific rules `suricata_custom_rules_x.rules`. If needed, this is updated within `suricata.yaml` for each instance.

2.4 Activity 4

The fourth activity is to reload all rules using `suricatasc -c reload-rules` in `notify` to fully automate this process. If successful, the deployment is finished; otherwise, the process returns for rebuilding.

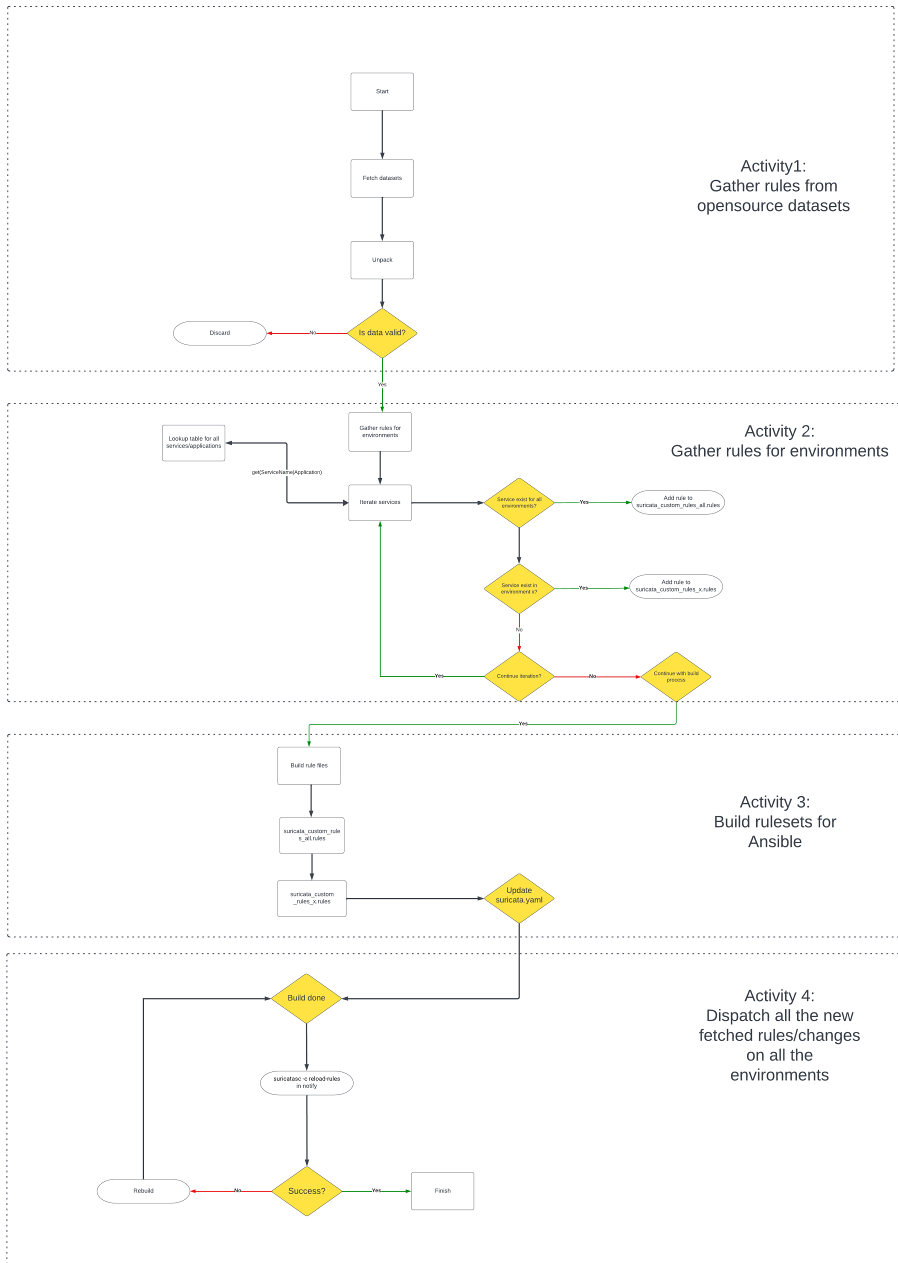

Figure 1: Framework structure with activities